

KUMAR HARSH

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Education

Indian Institute of Technology, Jodhpur

Bachelor of Technology in Bioengineering

Expected Graduation: 2027

Jodhpur, Rajasthan

Delhi Public School, Bokaro

Class 12 CBSE PCM (Percentage : 85.16%)

2022

Bokaro, Jharkhand

DAV Public School, Dhanbad

Class 10 CBSE (Percentage : 89.4%)

2020

Dahbad, Jharkhand

Projects and Experience

Sensor Fault Detection using Machine Learning Pipeline

Jan 2026 - Mar. 2026

- **Tech Stack:** Python, Flask, scikit-learn, XGBoost, Pandas, NumPy, HTML/CSS, Jinja2
- Developed an end-to-end machine learning pipeline to detect faulty wafer sensors using 500+ sensor features from manufacturing data.
- Trained and evaluated four ML models (Random Forest, Gradient Boosting, SVM, XGBoost) using GridSearchCV for hyperparameter tuning, achieving ~92–95% classification accuracy.
- Built a Flask-based prediction web application allowing users to upload CSV files and generate predictions for 1000+ sensor records within seconds.
- Implemented structured logging and custom exception handling to improve debugging efficiency and enhance pipeline reliability.

Smart Research Assistant

Jan 2026 - March 2026

- Tech Stack: Python, LangChain, Google Gemini API, FAISS, Tavily API, Streamlit, PyPDFLoader, RecursiveCharacterTextSplitter, Google Generative AI Embeddings, dotenv, Git
- Built a Gemini-powered multi-tool AI agent with RAG (FAISS), web search (Tavily), and conversational memory using LangChain
- Implemented PDF ingestion, chunking, embedding, and retrieval pipeline for context-aware responses
- Designed query routing logic and retry handling to improve accuracy and system robustness
- Developed a Streamlit-based chat interface with modular architecture and session memory

Personal Finance Tracker

Jan. 2026 - Mar. 2026

- Tech Stack: Node.js, Express.js, PostgreSQL, Knex.js, Passport.js, OpenAI API, SendGrid, Multer, HTML, CSS, JavaScript.
- Built a full-stack financial management platform with transaction tracking, budget monitoring, and analytics using Node.js, Express, and PostgreSQL
- Implemented secure authentication using Passport.js (JWT, Google OAuth) and integrated CSV import/export with file handling via Multer
- Developed AI-powered financial insights and anomaly detection using OpenAI API, along with automated email alerts via SendGrid.
- Designed RESTful APIs and a responsive frontend (HTML, CSS, JavaScript), and deployed the application on Render with production-ready setup

Technical Skills

Languages: Python, SQL (Basic), C++, JavaScript

Databases: MongoDB, SQL (MySQL/PostgreSQL - Basic)

Data Analysis & ML: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch

Data Visualization: Matplotlib, Seaborn, (Basic: Excel Charts / Power BI)

Web & APIs: Flask, REST APIs, React.js, Node.js

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux

Relevant Courses

- **Completed:** Machine Learning, Data Structures and Algorithms, Probability & Statistics, Stochastic Processes, Signal & System

Certificates

- * **Coursera:** Deep Learning Specialization, Machine Learning Specialization
- * **UDEMY:** DSA in C and C++

Achievements

- Solved 200+ DSA problems across LeetCode, GeeksforGeeks, and other coding platforms
- Secured 97.72 percentile in JEE Mains among 1.2 million candidates.

Extracurricular & Positions of Responsibility

Core Member, FRAME-X, IIT Jodhpur <i>Led videography and video editing for music videos and the Freshers’ video.</i>	2024 – 2025
Participated in volunteer roles during Varchas and Prometeo events <i>Volunteered under Assistant Heads to ensure the successful execution of the Fest.</i>	2023 – 2024